📑 **Final Project Report**

## 1. Project Objective

The objective of this project was to design a practical analytics solution that helps small restaurant owners make data-driven decisions. Specifically, the project aimed to:

- Analyze historical monthly sales data
- Identify long-term trends and seasonal patterns
- Forecast future sales
- Develop a management dashboard
- Generate actionable business recommendations

---

## 2. Dataset Overview

The dataset consists of more than 24 months of restaurant sales data from April 2024 to April 2026. Each observation represents total sales for a single month.

Additional analytical fields were created, including:

- Month-over-Month (MoM) Growth
- Trend Index
- Three-Month Moving Average
- Forecasted Sales
- Month Names

---

## 3. Methodology

### Data Preparation

The raw sales data was cleaned and standardized in Excel. Dates were organized in a consistent format and supporting analytical columns were added.

**Trend Analysis**

A line chart was created to visualize monthly sales performance over time.

**Moving Average**

A three-month moving average was calculated to reduce short-term volatility and reveal the underlying trend.

**Forecasting**

Excel's FORECAST.LINEAR() function was used to estimate future sales based on historical trends.

**Dashboard Development**

A professional dashboard was designed to present:

- Total Sales
- Average Monthly Sales
- Highest Sales Month
- Next Month Forecast
- Trend Chart
- Key Insights

---

## 4. Key Findings

- Total sales over the analysis period exceeded ¥29 million.
- Average monthly sales were approximately ¥1.28 million.
- Peak sales occurred in April 2025.
- Sales increased significantly during March and April in both 2025 and 2026.
- The recurring spring increase suggests seasonality related to sakura season, graduation events, and tourism.

- Forecast results indicate stable future sales around ¥1.2 million per month.

---

**5. Business Recommendations**

Based on the analysis, restaurant management should:

- Increase staffing during March and April to accommodate higher customer demand.
- Expand inventory and supplies ahead of peak seasonal periods.
- Develop targeted marketing campaigns during the spring season.
- Use forecasting results to improve budgeting and purchasing decisions.

---

**6. Skills Demonstrated**

This project demonstrates practical skills in:

- Microsoft Excel
- Data Cleaning
- Time-Series Analysis
- Moving Averages
- Forecasting
- Dashboard Design
- Business Interpretation
- Decision Support Systems

---

**7. Conclusion**

This project illustrates how historical sales data can be transformed into strategic business insights through practical analytics techniques. By combining data preparation, visualization, forecasting, and dashboard development, the project provides a scalable decision-support tool for small businesses seeking to improve operational planning and financial performance.

---